

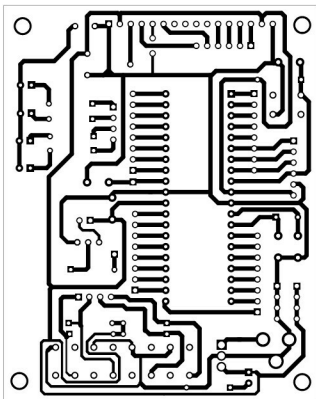


Fig. 7: Actual-size PCB layout of Arduino for ATmega32

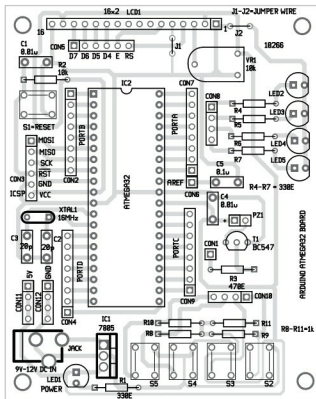


Fig. 8: Component layout of the PCB

Construction and testing

An actual-size, single-side PCB pattern of Arduino on ATmega32 circuit is shown in Fig. 7 and its component layout in Fig. 8. Assemble the circuit on the PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any error(s). Use proper IC base for the MCU. Test different

```
if defined(ISC2)
ISR(INT2_vect) {
if (intFunc[EXTERNAL_INT_2])
intFunc[EXTERNAL_INT_2]();
}
#endif
```

Open wiring.c in the same directory cores and add following after line number 201:

```
#if defined(TCCRO) && defined(WGM00) &&
defined(WGM01)
sbi (TCCRO, WGM00);
sbi (TCCRO, WGM01);
#endif
```

In the same directory open Wiring_analog.c and add following after line number 129:

```
#if defined(_AVR_ATmega32_)
case TIMER0:
sbi (TCCRO, COM01);
OCR0 = val; // set pwm duty
break;
#endif
```

In the same directory open Wiring_digital.c and add following after line number 98:

```
#if defined(TCCRO) && (COM01)
case TIMER0: cbi (TCCRO,
COM01); break;
#endif
```

In the same directory open wiring_private.h and add following after line number 56:

```
#elif defined(_AVR_ATmega1284_) ||
defined(_AVR_ATmega1284P_) || defined(_
AVR_ATmega644_) || defined(_AVR_
ATmega644A_) || defined(_AVR_
ATmega644P_) || defined(_AVR_
ATmega644FA_) || defined(_AVR_
ATmega32_)
```

These modifications are sufficient. You can now select Arduino ATmega32 under Tools > Board on Arduino IDE 1.0.5 as shown in Fig. 5. You may perform all the steps for ATmega32 on this IDE like any other Arduino board. But with this you cannot load the sketches without an external programmer because we have not loaded bootloader on ATmega32 yet.

However, without the bootloader, it is also advantageous as full memory space is available for sketches/application programs on ATmega32. If we use bootloader, it will consume 2k memory of ATmega32.

With this project, you can use Arduino IDE for various designing purposes using ATmega32 MCUs.

programs like digital read, analogue read, LCD display, serial display, tone, etc using ATmega32 for Arduino with all 32 GPIO pins.

The hex file generated (after compiling the sketch for Arduino board using ATmega32) on Arduino 1.0.5 IDE can be obtained by searching the build folder in C:// as shown in Fig. 6.

You may connect an external programmer to CON3 in Arduino ATmega32 board to burn the hex file into ATmega32.

You can also change any future version of Arduino for ATmega32 once you understand how to change the corresponding files in that version.

EFY note. 1. The relevant files of this article are included in this month's EFY DVD and are also available for free download at source.efymag.com

2. Your code may not match the line numbers mentioned in this article. **EFY**



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